Response to reviews: 2025-155

We thank the editor, associate editor and reviewers for their careful assessment of our paper. What follows is a point by point response to the reviewers' comments. The original reviewer comment is in normal font and *our response is in italic text*.

Reviewer 1

Recommendation

My overall impression from the package is positive. However, several things can be done to improve the paper and the package. Please see a brief summary below. More illustrations of using the code along with demo files should be provided. The authors should not oversell their package by saying that it can help evaluate clustering methods. If they insist on this package's capability, perhaps, they need to outline how exactly they plan to achieve that. I agree that it can be helpful to simulate an interesting dataset, but its clustering complexity will be unknown. Grammar/professional presentation needs to be checked/improved. At places, the manuscript and package are slopy.

General

The manuscript "cardinalR: Generating Interesting High-Dimensional Data Structures" presents a wide variety of methods to simulate datasets of various shapes. While these capabilities are undoubtfully useful, the presentation of the material should be somewhat changed in my opinion. In particular, the authors need to provide more examples that would stimulate readers' interest. Also, the authors should be careful with claiming that their package can be helpful for studying cluster analysis methods. To summarize, I think the package is useful, but presentation of the material should be improved.

Detailed comments

1. The authors claim that the package helps study clustering and dimension reduction algorithms. While I have no concerns with regard to dimension reduction algorithms, I would like to encourage the authors to provide more details regarding how the package can be used for studying cluster analysis procedures. This is very much related to Figure~1 provided by the authors. Having just shape generator routines is not sufficient to claim that the package is useful for evaluating clustering algorithms. The clustering complexity of a particular dataset is defined not just by the geometry of clusters themselves, but also by their proximity to and interaction with each other. For example, no matter what cluster shapes a user chooses, if the clusters are far enough from each other, even a simple k-means algorithm will find the perfect cluster assignment. In my opinion, the authors should say no more than that their package allows simulating clusters of different shapes. At the same time, this is not particularly useful for studying the systematic performance of clustering algorithms.
2. This point is closely related to the previous one. What is a cluster according to the authors' definition? There are many alternative definitions. If the authors want to claim that their package can help evaluate clustering algorithms based on datasets simulated from their package, there should be a discussion regarding what constitutes a cluster.

Please see some useful references below:

- Hennig, C. (2015) What are the true clusters?, Pattern Recognition Letters, 6, 53-62.
- Maitra, R. and Melnykov, V. (2010) Simulating data to study performance of finite mixture modeling and clustering algorithms, Journal of Computational and Graphical Statistics, 2(19), 354-376

3. page 2: "A full list of available shape generators are available…"

- "IS available"; also, avoid using "available" twice

We have corrected the sentence.

4. Don't make space before "where" in the beginning of a line (see, for examples, page 6)

We have fixed this.

5. Please check the manuscript for English and typos.

We have checked.

Code

1. I think that the number of examples is insufficient. Many more code-related illustrations motivating the use of the package need to be provided, preferably with corresponding demo-files. These should not be massive, but can illustrate different aspects of the package use.

2. Please make sure that all help files and messages provided by the package are informative, professional, and grammatically correct; for example, remove "!!!" and correct the comments below "Data generation completed successfully!!! Multiple clusters generation completed successfully!!!"

Reviewer 2

Location notation. Throughout this report, locations in the manuscript are referenced as Pn/T– M–B, where Pn is the page number and T, M, B stand for top, mid, and bottom third of the page, respectively. For example, P3/T means the top third of page 3.

Overview

This paper presents `cardinalR`, an R package for generating synthetic high-dimensional datasets with controlled geometric properties. The package provides a library of shape generators (branching structures, cones, cubes, pyramids, spheres, spirals, trefoil knots, and others) along with a main orchestration function, `gen_multicluster()`, that combines multiple shapes into a single multi cluster dataset with user-specified centroids, scaling, rotation, and background noise. The contribution is computational. The package fills a genuine gap in the R ecosystem: while existing packages (`geozoo`, `snedata`, `mlbench`) provide individual shape generators, none offers a unified framework for composing multi-cluster scenes with heterogeneous geometries in a controlled high-dimensional setting.

Strengths. The compositional framework implemented by `gen_multicluster()` is the most original and useful contribution of this work. The ability to combine arbitrary shapes with inde pendent control of location, scale, and rotation is genuinely valuable for benchmarking nonlinear dimension reduction and clustering methods. The variety of generators is broad, and the mathematical documentation of each generator is more detailed than what is available in competing packages. The package is available on CRAN and GitHub with a pkgdown website.

Weaknesses. The paper has several significant issues that need to be addressed before publication. First, the paper is too long and reads like a catalog. Second, the Application section (Section 4) is weak. The example uses only 4 dimensions, which is unrepresentative for a package that emphasizes "high-dimensional" data. Third, there are issue with the companion script and other coding issues

Recommendation. I recommend **major revision**. The core contribution—the compositional framework and the original generators—has sufficient value for publication in The R Journal. How ever, the paper requires substantial restructuring. The issues detailed below should be addressed thoroughly.

Major Issues

M1. Section 2 (Usage) should be eliminated and its content redistributed. Section 2 (P2–P3) contains a single code example for `gen_multicluster()`, Table 1 (arguments), and Figure 1 (workflow diagram). This material does not stand on its own. Removing Section 2 would reduce the paper length (currently 23 pages, exceeding the R Journal's 20-page guide line) without any loss of content.

M2. Section 3 (Implementation) is too long and does not distinguish original contributions from wrappers. Section 3 spans approximately 13 pages and treats every generator with the same level of detail, regardless of whether it is a genuinely new contribution or a thin wrapper around an existing function. For instance, `gen_mobius()` is declared as a wrapper around `geozoo::mobius()`, `gen_gridcube()` wraps `geozoo::cube.solid.grid()`, etc. These should be acknowledged briefly (a line or two each), with their full details relegated to the package documentation or a supplementary file. The paper body should reserve detailed exposition for generators that are genuinely new: the branching families, pyramids, spirals, trefoil knots, the hole-generation mechanism, and above all `gen_multicluster()` itself. This re structuring would simultaneously solve the length problem, reduce the catalog-like monotony of the current Section 3, and make the paper's actual contribution much clearer.

M3. Section 3 contains numerous undefined parameters, notational inconsistencies, and apparent errors. A non-exhaustive list:

- P5/T: In `gen_linearbranches`, δ (jitter) is never defined and is not a function argument. Same for smin, smax.

We have fixed this.

- P6/B: The parameter ratio is described as "radius tip to radius base ratio" (Table 3) but is used as an absolute value rmin, not as a ratio.
- P8/T: In `gen_pyrrect`, coordinates X_1 and X_3 are given the same distribution?
- P9/B–P10/T: For `gen_unifsphere`, the text first says points are on the surface of a sphere, then states it produces a "solid ball." These are contradictory.
- P10/B: In `gen_clusteredspheres`, the parameters n_1, n_2, r_1, r_2 used in the text do not match the function signature (n, k, r, loc) from Table 2.
- P15/T: In `gen_noisedims`, the sign-alternation of odd dimensions is described as avoiding "directional drift," does it make sense for zero-mean Gaussian variables?

We agree that for zero-mean Gaussian noise, sign alternation does not affect directional drift. It is a simple transformation that preserves the distribution and independence. We have revised the text to clarify this.

- P15/M: `gen_wavydims1` generates $X_j = \alpha_j \theta + \epsilon_j$, which is linear—not "wavy."
- Throughout: the notation for random variables oscillates among X_j , x_j , X_{ij} , and R column names x_1, x_2 without a declared convention.
- There are other issues like the previous. The authors are suggested to make a systematic scan across the paper and scripts submitted.

M4. Section 4 (Application) does not adequately demonstrate the package and draws unsupported conclusions. Several specific problems:

- The example dataset uses only 4 dimensions, no rotation, and no noise dimensions. For a package whose title emphasizes high-dimensional data, this is unrepresentative.
- The six DR methods (P17/M) are applied without specifying hyperparameters. The companion script reveals some parameters via the file names of pre-computed RDS files (e.g., perplexity 30 for t-SNE, k-nearest-neighbors 5 for PHATE), but these are never stated in the paper. This makes the results non-reproducible from the paper alone.
- The HBE metric (P17/M) is not defined in the paper.
- P17/B: The author claim that t-SNE shows "high preservation of both local and global structures". Please check it, I doubt it is correct.

We have corrected the statement.

- P17/B: The text refers to "six clusters" in the UMAP layout, but the dataset has five clusters.

We have revised the wording.

- P18/M: Model-based clustering uses the "EII" parameterization (spherical, equal vol ume). Since the clusters are non-spherical by construction, this choice severely penalizes the method. Either justify this choice or include a flexible parameterization (e.g., "VVV").
- The clustering evaluation does not compute any external validity index (ARI, NMI) against the known ground truth. This is a missed opportunity: the main advantage of synthetic data with known labels is precisely that one can measure recovery of the true partition. I recommend a substantially revised Section 4 that: uses a higher-dimensional scenario; reports DR hyperparameters; defines or replaces HBE; includes standard external cluster validity measures; and demonstrates distinctive features of the package (rotation, noise dimensions, holes).

M5. Package API mismatch and missing changelog. The paper (Table 1, P3) uses the argument `is_bkg` for `gen_multicluster()`, while the current pkgdown vignette (version 1.0.5) uses `add_bkg`. This discrepancy indicates either the paper describes an outdated API or the package changed after the paper was written. The paper does not state which version of the package was used, which is recommended by R Journal guidelines. Additionally, the NEWS.md file is completely empty across all six CRAN releases—no description of changes, bug fixes, or breaking changes is available to users.

M6. Companion script is not self-contained. The submitted R script (paper-cardinalR.R) cannot be executed independently. It depends on `scripts/additional_functions.R`, which is not provided. The DR results and cluster statistics are loaded from pre-computed RDS files hosted on GitHub rather than being computed within the script. The reader can regenerate the figures from those pre-computed objects but cannot verify how they were obtained, with which packages, or with which parameters. Full reproducibility requires that either the complete pipeline be included in the submission or clearly documented in a publicly accessible repository with all dependencies explicit.

We have updated the script to ensure it can be executed independently.

All dimension reduction results are precomputed and provided as RDS files in the repository. The full code used to generate these results, including parameter settings and package versions, is available in the repository and allows complete reproduction of all analyses.

Minor Issues

m1. P2/M: The code example for `gen_multicluster()` is truncated—the closing of the `loc` matrix and the function call are missing. The reader cannot execute the code as printed.

We have corrected this.

m2. P2/T: The paper states the package is "available on CRAN" but does not indicate the version used for the paper, as recommended by R Journal guidelines.

We have added the CRAN version.

m3. P3, Table 1: The type of `is_bkg` is listed as "boolean"; the standard R term is "logical." The type of rotation is listed as "numeric (list)"—a more precise description would be "list of numeric matrices."

We have corrected the terminology.

m4. P7/T: For `gen_unifcube`, the removal of cube vertices after generation is stated without motivation. Does this alter the sampling distribution and should be justified?

We have clarified that the removed points correspond to deterministic cube vertices; their removal does not affect the uniform sampling distribution.

m5. P7/B: The noise variance in `gen_longlinear` scales as $(0.03n)^2$. For $n = 1000$, the standard deviation is 30, which can dominate the signal depending on a_j and b_j. Is it intentional?

m6. P8/M: The range argument of `gen_quadratic` and `gen_cubic` does not appear in Tables 2 or 3.

m7. P9/T: The height distribution for `gen_pyrstar` is uniform, whereas for the other pyramids it is exponential. The difference is not motivated.

In `gen_pyrstar`, we use a uniform height distribution to spread points evenly from the base to the tip, so the star shape is clearly visible. In the other pyramid generators, we use an exponential distribution, which places more points near the tip. We have clarified this in the manuscript.

m8. P9/M: For `gen_pyrfrac` (chaos game), it is not stated whether a burn-in period is used to discard transient iterates, which is standard practice.

No burn-in period is currently used; however, due to the contractive nature of the midpoint update, the influence of the initial point diminishes rapidly. We have clarified this in the manuscript.

m9. P9/B: The naming of the Sierpinski-like pyramid is inconsistent: the function is `gen_pyrfrac`, but Figure 4 and the text refer to it as "pyrholes."

We have corrected the naming inconsistency.

m10. P11/M: In `gen_swissroll`, the parameter w is described as "Vertical variation" (scalar) in Table 3 but used as a two-component vector (w_1, w_2) in the formula.

m11. P17/M: The cluster positions are generated with `geozoo::simplex(p=4)$points` without explaining what this produces (5 vertices of a 4-D simplex). The dimensionality of the example (4-D) and the relationship between the simplex vertices and the number of clusters are left entirely implicit.

m12. P18/M: "Model-based clustering performed the 'EII' covariance structure" is grammatically incorrect; it should be "was performed using" or "employed."

We have revised the wording.

m13. P1/M: “High-dimensional data is characterized”—“data” takes a plural verb in formal academic English (“data are”).

We have revised the wording.

m14. P1/B: “resource researchers with synthetic datasets”—“resource” used as a verb is non-standard; “provide” would be appropriate.

We have revised the wording.

m15. P5/M: Several sentences introducing mathematical formulas do not close grammatically before the displayed equation.

We have fixed this.

m16. P18/T, Figure 9 caption: “PAHTE” is a typo for “PHATE.”

We have corrected this.

m17. The companion script loads `library(tidyverse)` (line 21), which is discouraged for reproducibility scripts. Only the actually needed packages (`dplyr`, `tidyr`, `ggplot2`) should be loaded.

We have updated the companion script to load only the required packages.

m18. P20: The Boehmke and Greenwell (2019) reference for k-means is an introductory textbook. A standard reference (e.g., Hartigan and Wong, 1979) would be more appropriate.

We have replaced the textbook citation with the standard reference Hartigan and Wong (1979) for k-means clustering.

m19. P20: The Devroye (1986) bibliographic entry appears malformed (“originally published with” is truncated).

We have corrected this.

m20. P21: The Amid and Warmuth (2019) TriMAP reference is listed as an ArXiv preprint. If a published version exists, it should be cited instead.

We have updated the reference to the latest version of the TriMAP arXiv preprint (Amid & Warmuth, 2019). We checked and TriMAP is only available as an arXiv preprint, with no published journal or conference version. We therefore keep the arXiv citation.

m21. P21: The Melville (2025) `snedata` reference cites a GitHub package at version 0.0.0.9001 (development version, not on CRAN). This should be noted explicitly.

We have updated the reference to clarify that snedata is a GitHub development version (0.0.0.9001), not a CRAN release, and we include the commit hash and repository URL.

m22. t-SNE is used throughout Section 4 but van der Maaten and Hinton (2008) is never cited. It is the only DR method in the comparison without a reference.

We have added the missing citation for t-SNE (van der Maaten and Hinton, 2008) in Section 4.

m23. The R packages used to run the DR methods in Section 4 (`Rtsne`? `uwot`? Python via `reticulate`?) are never disclosed. The R Journal guidelines require that all packages used be declared.

We have added these.

m24. `scikit-learn` (Pedregosa et al., 2011) is not cited despite several generators (S-curve, Swiss roll) originating from that ecosystem.

We have added this.

m25. P21: The Gamage et al. reference is listed as 2019 but the ArXiv identifier (2506.22051) indicates a 2025 submission date.

We have corrected this.

m26. P22: The Witten (1985) reference ("Non-commutative geometry and knot theory") appears to be incorrect. Check the year of publication.

We have corrected this.

m27. P12: Laurent (2024) is a blog post—a fragile source for a peer-reviewed paper.

We have replaced this blog reference with appropriate peer-reviewed sources.

m28. P18 caption of Figure 9: "PAHTE" should be "PHATE."

We have corrected this.

m29. P19 Figure 11 caption: both rows of panels are labeled a1–a3; the second row should be b1–b3.

We have corrected this.